



IronKey

The IronKey is marketed as the most secure USB drive in the world. The drive is lined with a small amount of explosive, to self-destruct on intrusion

Secure your portable device

Prevent your data from being abused in case of theft, loss or casual snooping on your portable devices.

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Portable memory has become a mainstay of modern computer usage. Executives, employees, government officials and individual users have a range of reasons to protect the data on their portable devices. Two basic security measures include password protecting the

PicoCrypt is a small encryption and decryption tool. It can be used to encrypt any kind of file. It is a small programme (16 KB) that runs without an installation, and can be carried around on a USB drive. When used to encrypt a file, the file will be visible, but not readable. An image or a text file will appear garbled or as gibberish. Some kinds of extensions cannot be opened at all. However, the thumbnail of

the document will be visible if Windows has the thumbnail saved before the file was encrypted. Also, users can delete the file from the system if they want to.

To encrypt a file, simple

drag and drop it into PicoCrypt. A window will ask you if you want to encrypt or decrypt the file. Click on encrypt. You will be prompted for a password, and a confirmation. If you drag and drop multiple files into PicoCrypt, you will be prompted to enter a password and confirm it for each of the files dragged into PicoCrypt. To view the files again, start up PicoCrypt, and drag and drop the files you want to decrypt, then enter the passwords for each of them in turn. Now the files can be opened and edited on programmes.

Another simple way to protect files is to put them in an encrypted archive. This is essentially the same as the above method, but the extension is changed to archive

format, such as a ZIP file. We will be using a free and open source archiving software called 7zip. 7zip stores archives in the .7z format by default, but also works with BZip2, GZip, TAR and ZIP formats. WinRAR is a great software for handling archive files, but you have to purchase the full version before you can password protect your archives. Once you install 7zip, the archiving operations show up in the context menu when you right click on a file or folder you want to archive. Right-click and select "add to archive". In the Encryption field, enter a password, and enter it again. If you check the "Show password" checkbox, you don't need to enter the password twice. In the Archive Format drop down menu, select the archive format you want to use. 7zip portable and one of the more obscure formats, will discourage casual snoopers. If anyone wants to access the contents of the archive, they will need the password. This is a great way of sharing files over public hosting services, such as yousendit or rapidshare.

People can still delete the archive if they want to. To circumvent this, archives can be hidden inside other kinds of files, such as an image or a text file. To do this, once the archive is created, you will have to navigate to the folder using the command prompt. Go to Start>Run>cmd and hit [Enter]. First, copy an

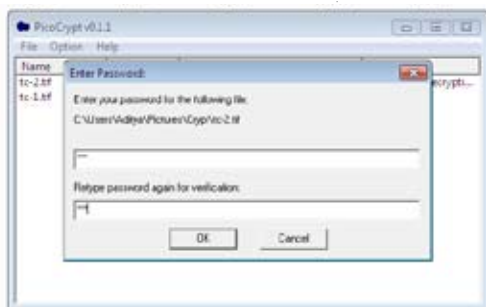
image file to the same folder as the archive. Then use `cd or cd . .` to navigate to the relevant folder. Once there, enter the following command.

```
copy /b image.jpg + archive.7z image2.jpg
```

The image2.jpg will be the image file with the hidden archive. To extract the archive, change the extension of the file from image2.jpg to archive.7z. Note that you need to remember the format the archive was originally in. Using a .zip extension for a .7z archive won't work. The image2.jpg file can be opened in image viewers, and even in image manipulation programs. The archive will get corrupted only if the image file is overwritten with changes.

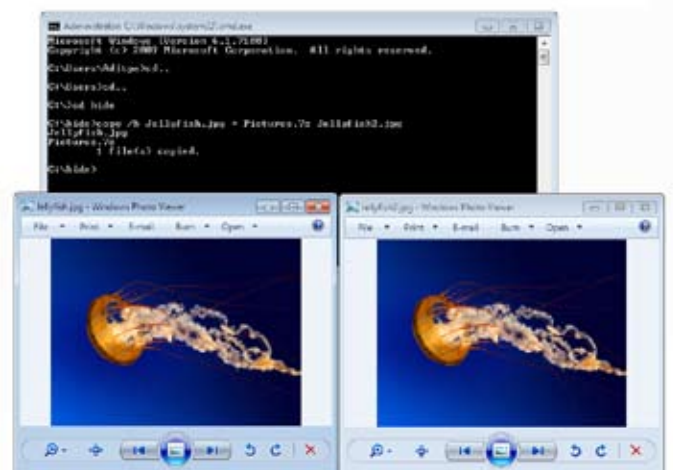
Sometimes it is necessary to hide or lock away entire folders. When working regularly with files, the above methods are not of much help, and are pretty laborious.

TrueCrypt is a great open source encryption software for such purposes. When installing TrueCrypt, there are two options available. A disk installation will install the software on the hard drive. An extraction, will however place the software on an external drive, and can be run from there. This is essentially a portable version of TrueCrypt, but this version is limited in capability. After you encrypt



Simple encryption and decryption of single files

data, and encrypting it. Many portable device manufacturers, such as Buffalo Systems or Kingston, have versions of portable memory devices that offer in-built encryption and/or password protection. This is achieved through the agency of software that comes pre-installed on the portable device itself. Such devices are marginally more secure than the methods described below. These software are pretty straightforward to setup and use. Invariably, such software allow users to protect a section of the data stored on the portable device, or all the data stored on the device. The methods described below will allow you to use some simple tools for effectively disallowing access to your files.



The image file, the archive to be hidden, the command line code, and the two images opened side by side